

$$J=3 \quad p(x) = \begin{cases} \frac{e^{-\frac{x}{\theta}}}{\theta}, & x \geq 0 \\ 0, & x < 0 \end{cases}, \theta > 0$$

$$n=3, \quad \tilde{\theta}_1 = \bar{x}, \quad \tilde{\theta}_2 = x_{(2)}$$

a) 1) $\tilde{\theta}_1 = \bar{x}$, где эквив. распр. $z = \theta$
 $\mathcal{M}\tilde{\theta}_1 = \mathcal{M}\bar{x} = \mathcal{M}\left(\frac{1}{n} \sum x_i\right) = \frac{1}{n} n \mathcal{M}z = \mathcal{M}z = \theta \Rightarrow$ несмещ.

2) $\tilde{\theta}_2 = x_{(2)}$

$$F(x) = \int_0^x e^{-\frac{t}{\theta}} \cdot \frac{1}{\theta} dt = 1 - e^{-\frac{x}{\theta}}, \quad x \geq 0$$

$$n=3, k=2:$$

$$\begin{aligned} \mathcal{L}(x) &= 3 \cdot e^{-\frac{x}{\theta}} \cdot \frac{1}{\theta} \cdot C_2 \left(1 - 1 + e^{-\frac{x}{\theta}}\right)^1 \left(1 - e^{-\frac{x}{\theta}}\right)^1 = \\ &= 3 \cdot e^{-\frac{x}{\theta}} \cdot \frac{1}{\theta} \cdot 2 \cdot e^{-\frac{x}{\theta}} (1 - e^{-\frac{x}{\theta}}) = \\ &= \frac{6}{\theta} \cdot e^{-\frac{2x}{\theta}} (1 - e^{-\frac{x}{\theta}}) \end{aligned}$$

$$\mathcal{M}\tilde{\theta}_2 = \int_0^{+\infty} x \cdot e^{-\frac{2x}{\theta}} \cdot \frac{6}{\theta} (1 - e^{-\frac{x}{\theta}}) dx = \frac{5}{6} \theta - \text{смещ.}$$

$$\Rightarrow \tilde{\theta}_2' = \frac{6}{5} \tilde{\theta}_2 \Rightarrow \text{несмещ.}$$

б) 1) $\mathcal{D}\tilde{\theta}_1 = \mathcal{D}\left(\frac{1}{n} \sum x_i\right) = \frac{1}{n^2} n \cdot \mathcal{D}z = \frac{\theta^2}{3}$

2) $\mathcal{M}\tilde{\theta}_2 = \frac{5}{6} \theta$

$$\mathcal{M}\tilde{\theta}_2^2 = \mathcal{M}x_{(2)}^2 = \frac{6}{\theta} \int_0^{+\infty} (x^2 \cdot e^{-\frac{2x}{\theta}} - x^2 \cdot e^{-\frac{3x}{\theta}}) dx \equiv$$

1) $u = x^2 \quad dv = e^{-\frac{2x}{\theta}} dx$
 $du = 2x \quad v = \frac{\theta}{2} \cdot e^{-\frac{2x}{\theta}}$

2) $u = x^2 \quad dv = e^{-\frac{3x}{\theta}} dx$
 $du = 2x \quad v = -\frac{\theta}{3} \cdot e^{-\frac{3x}{\theta}}$

2 раз по частям : сначала 1) потом 2)

$$\begin{aligned} &= \frac{6}{\theta} \left(\left(-\frac{\theta}{2} x^2 \cdot e^{-\frac{x}{\theta}} \right) \Big|_0^{+\infty} + \left(\frac{\theta}{3} x^2 \cdot e^{-\frac{x}{\theta}} \right) \Big|_0^{+\infty} + \right. \\ &+ 2 \cdot \frac{\theta}{2} \int_0^{+\infty} x \cdot e^{-\frac{x}{\theta}} dx - 2 \cdot \frac{\theta}{3} \int_0^{+\infty} x \cdot e^{-\frac{x}{\theta}} dx \Big) = \\ &= \frac{6}{\theta} \left(\frac{\theta^3}{4} - \frac{2\theta^3}{27} \right) = \frac{13}{18} \theta^2 \end{aligned}$$

$$D\tilde{\theta}_2 = M\tilde{\theta}_2^2 - (M\tilde{\theta}_2)^2 = \frac{13}{36} \theta^2$$

$$D\tilde{\theta}_1 = \frac{36}{25} \cdot \frac{13}{36} \cdot \theta^2 = \frac{13\theta^2}{25}$$

$$\cancel{D\tilde{\theta}_1} \quad D\tilde{\theta}_1 < D\tilde{\theta}_2 \Rightarrow \tilde{\theta}_1 - \text{эффективнее}$$

$$\begin{aligned} \text{б) } I(\theta) &= M \left[\left(\frac{\partial \ln g(x, \theta)}{\partial \theta} \right)^2 \right] = M \left[\left(\frac{\partial \ln(e^{-x/\theta}/\theta)}{\partial \theta} \right)^2 \right] = \\ &= M \left[\left(\frac{x}{\theta^2} - \frac{1}{\theta} \right)^2 \right] = \frac{1}{\theta^2} > 0 \quad \text{и } \theta \in C[0; +\infty) \Rightarrow \end{aligned}$$

1) $g(x, \theta) = p(x, \theta)$ - непрерывная функция на $(0; +\infty)$

$$2) \frac{\partial}{\partial \theta} \int_{-\infty}^{+\infty} dF(x, \theta) = \frac{\partial}{\partial \theta} \int_0^{+\infty} e^{-x/\theta} \cdot \frac{1}{\theta} dx = 0$$

$$\begin{aligned} \int_0^{+\infty} \frac{\partial}{\partial \theta} \left(e^{-x/\theta} \cdot \frac{1}{\theta} \right) dx &= \int_0^{+\infty} \left(\frac{x \cdot e^{-x/\theta}}{\theta^3} - \frac{e^{-x/\theta}}{\theta^2} \right) dx = \\ &= \frac{1}{\theta^3} \int_0^{+\infty} x \cdot e^{-x/\theta} dx - \frac{1}{\theta^2} \int_0^{+\infty} e^{-x/\theta} dx = 0 \end{aligned}$$

$$0=0 \Rightarrow$$

$\Rightarrow$ модель регулярна

$$D\tilde{\theta} = \frac{\theta^2}{3} - \text{оп. на } \forall \text{ компакте из } (0; +\infty)$$

$$\forall \theta \in [a, b] \subset (0; +\infty) \hookrightarrow \frac{\theta^2}{3} \leq \frac{b^2}{3}$$

$\Rightarrow$ по дост. усл. регулярна оценка $\tilde{\theta}_1$

$$D\tilde{\theta}_2 = \frac{13}{25} \theta^2 - \text{оп. на } \forall \text{ компакте из } (0; +\infty) \Rightarrow$$

$\Rightarrow$ по дост. усл. $\tilde{\theta}_2$ - регулярна

→ можем применить нер-во Крамера-Рао:

$$\forall \theta \in \Theta \quad D\tilde{\theta}_1 \geq \frac{1}{nI(\theta)} = \frac{\theta^2}{3}$$

$$D\tilde{\theta}_1 = \frac{\theta^2}{3} = \frac{\theta^2}{3} \Rightarrow \text{по дост. усл. эффе.}$$

→ т.к. $\exists!$ эффе. оценка, то $\tilde{\theta}_1$ - не будет эффе. $\tilde{\theta}_2$ - не будет эффе.

$$n=4$$

$$\theta = a$$

$$b = 1 - 2\theta$$

$$a \int_{-1}^1 dx + \frac{b}{2} + \frac{b}{2} = 1 \quad (\text{т.к. } \int_{-1}^1 p dx = 1) \Rightarrow$$

$$\Rightarrow b = 1 - 2\theta$$

$$\mu_Z = \int_{-1}^1 x \cdot \theta dx + \frac{2 \cdot (1 - 2\theta)}{2} + \frac{0 \cdot (1 - 2\theta)}{2} = 1 - 2\theta$$

$$\mu_{Z^2} = \int_{-1}^1 x^2 \cdot p(x) dx + \frac{4(1 - 2\theta)}{2} = 2 - \frac{10}{3}\theta$$

$$D_Z = \mu_{Z^2} - (\mu_Z)^2 = 1 + \frac{2}{3}\theta - 4\theta^2$$

ОММ

$$\hat{\mu}_1 = \tilde{\mu}_1 = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x} \Rightarrow 1 - 2\theta = \bar{x} \Rightarrow$$

$$\Rightarrow \tilde{\theta}_1 = \frac{1 - \bar{x}}{2}$$

$$\mu_{\tilde{\theta}_1} = \frac{1}{2} - \frac{\mu_{\bar{x}}}{2} = \frac{1}{2} - \frac{1}{2}(1 - 2\theta) = \theta \Rightarrow \text{несмещ.}$$

$$D\tilde{\theta}_1 = D\left[\frac{1}{2} - \frac{\bar{x}}{2}\right] = \frac{1}{n} \cdot \frac{1}{n^2} \cdot n \cdot D_Z = \frac{1}{4n} \left(1 + \frac{2}{3}\theta - 4\theta^2\right) \xrightarrow{n \rightarrow \infty} 0$$

→ состоятельна по дост. условию.

ОМП

$$L(\theta) = \prod_{i=1}^n p(x_i, \theta) = (1-2\theta)^m \cdot \theta^{n-m}$$

$$\ln L = m \cdot \ln(1-2\theta) + (n-m) \cdot \ln \theta$$

$$(\ln L)'_{\theta} = -\frac{2m}{1-2\theta} + \frac{n-m}{\theta} = 0$$

$$(\ln L)''_{\theta} = -2m \cdot 2 \cdot \frac{1}{(1-2\theta)^2} + (-1)(n-m) \cdot \frac{1}{\theta^2} < 0 \Rightarrow \theta \rightarrow \max$$

$$\frac{2m}{1-2\theta} = \frac{n-m}{\theta}$$

$$2m \cdot \theta = \frac{n-m}{2\theta} \cdot (1-2\theta)$$

$$n - 2n\theta - m + 2m\theta - 2m\theta = 0$$

$$\tilde{\theta}_2 = \frac{n-m}{2n} = \frac{1}{2}(1-p)$$

$$\begin{aligned} \mu p &= p \\ \sigma^2 p &= \frac{p(1-p)}{n} \end{aligned}$$

$$\begin{aligned} \mu \tilde{\theta}_2 &= M\left[\frac{1}{2} - \frac{1}{2}p\right] = \frac{1}{2} - \frac{1}{2}Mp = \\ &= \frac{1}{2} - \frac{1}{2}(1-2\theta) = \theta - \end{aligned}$$

$$\sigma^2 \tilde{\theta}_2 = \frac{1}{n} \sigma^2 p = \frac{1}{n} \cdot \frac{(1-2\theta) \cdot 2\theta}{n} \xrightarrow{n \rightarrow \infty} 0 \Rightarrow \text{несмещ., точ. пог. уел.}$$

c) 1) $\tilde{\theta}_1: \frac{\partial}{\partial \theta} \int_{-\infty}^{\infty} p(x) dx = 0$

$$\int_{-1}^1 \frac{\partial}{\partial \theta} p(x, \theta) dx + \frac{\partial}{\partial \theta} (1-2\theta) = 2-2=0$$

$$\begin{aligned} I(\theta) &= \int_{-1}^1 \left(\frac{\partial \ln p(x, \theta)}{\partial \theta} \right)^2 p(x, \theta) dx + \left(\frac{\partial \ln p(\theta)}{\partial \theta} \right)^2 p(\theta) = \\ &= \int_{-1}^1 \left(\frac{1}{\theta} \right)^2 \theta dx + \frac{4}{(1-2\theta)^2} (1-2\theta) = \frac{2}{\theta} + \frac{4}{1-2\theta} = \end{aligned}$$

$$= \frac{2}{\theta(1-2\theta)}, \theta \in (0; \frac{1}{2}), I(\theta) > 0 \text{ и } I(\theta) - \text{непр. на } A \Rightarrow \text{модель регулярна}$$

1. Модель регулярна
2. Оценка несмещ.
3. $D\hat{\theta}_1$ опр. на $\forall$ компакте $из (0; \frac{1}{2})$

→ по дост. усл. оценка регулярна.

⇒ можно применить нер-во Крамера-Рао

$$D\hat{\theta}_1 \geq \frac{1}{n \cdot I(\theta)} = \frac{\theta(1-2\theta)}{2n}$$

$$\frac{1}{2n} \left(1 + \frac{2}{3}\theta - 4\theta^2 \right) \geq \frac{\theta}{2n} \cdot (1-2\theta)$$

$$1 + \frac{2}{3}\theta > 2\theta$$

2) $\hat{\theta}_2$: 1) модель регулярна

2) несмещ.

3) $D\hat{\theta}_2 = \frac{(1-2\theta)\theta}{2n}$ опр. на $\forall$ компакте $из (0; \frac{1}{2})$

⇒ нер-во Крамера-Рао:

$$D\hat{\theta}_2 \geq \frac{1}{n \cdot I(\theta)} = \frac{\theta(1-2\theta)}{2n}$$

$$\frac{\theta(1-2\theta)}{2n} = \frac{\theta(1-2\theta)}{2n} \rightarrow$$

⇒ по дост. усл. данная оценка

то $\hat{\theta}_1$ не эффективна, т.к. $\exists!$ эф. оценка